

Scientific Computation

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Nov. 18, 2025

Lecture 28 - Composite and Adaptive Rules

Quadrature Summary

We have looked at three basic schemes with the following quadrature rules and expressions for error:

Quadrature	Integral	Error
Midpoint	$(b-a)f(m)$	$\frac{f''(\eta)}{24}(b-a)^3$
Trapezoidal	$\frac{(b-a)}{2}[f(a) + f(b)]$	$-\frac{f''(\eta)}{12}(b-a)^3$
Simpson	$\frac{(b-a)}{6}[f(a) + 4f(m) + f(b)]$	$-\frac{f^{(4)}(\eta)}{90}\left(\frac{b-a}{2}\right)^5$

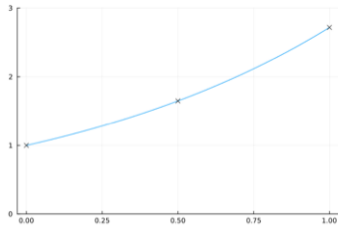
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Integrating the Exponent

Example 1: Compute $\int_0^1 e^x dx = e - 1 \approx 1.71828\dots$ once more, this time with

J_0

Simpson's rule.



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Lecture Outline

- Composite rules
- Accuracy of composite rules
- Adaptive Simpson's rule

Interval Sizes

The good news appears to be that increasing the degree of the interpolating polynomial decreases error. But are there other strategies?

We also saw last class that for the simple integral

$$\int_a^b e^x dx,$$

$$a = 0$$

$$b \in (0, 1, 1)$$

a smaller value of $b - a$ leads to smaller error. This empirical result matches the prediction of our error terms, which are $O((b - a)^{\mathfrak{K}})$, where $\mathfrak{K}$ depends on the rule.

Composite Rules

We will use the following strategy:

→ subdivide $[a, b]$ into $n > 1$ subintervals

→ $h = \frac{b-a}{r}$ length of each subinterval

→ $t_i = a + ih$, $i = 0, 1, \dots, r$

$t_0 = a$, $t_r = b$

$$I_f \triangleq \int_a^b f(x) dx = \sum_{i=1}^r \int_{t_{i-1}}^{t_i} f(x) dx$$

Composite Trapezoidal Rule

Applying the trapezoidal rule to each $[t_{i-1}, t_i]$ for $i = 1, \dots, r$ we get:

$$\int_{t_{i-1}}^{t_i} f(x) dx \approx \frac{t_i - t_{i-1}}{2} [f(t_{i-1}) + f(t_i)] = \frac{h}{2} [f(t_{i-1}) + f(t_i)]$$

$$\Rightarrow \int_a^b f(x) dx = \sum_{i=1}^r \int_{t_{i-1}}^{t_i} f(x) dx \approx \frac{h}{2} \sum_{i=1}^r [f(t_{i-1}) + f(t_i)]$$

$$= \frac{h}{2} \left(\sum_{i=1}^r f(t_{i-1}) + \sum_{i=1}^r f(t_i) \right)$$

$$= \frac{h}{2} (f(t_0) + f(t_1) + \dots + f(t_{r-1}))$$

$$+ \frac{h}{2} (f(t_1) + \dots + f(t_{r-1}) + f(t_r))$$

$$= \frac{h}{2} [f(a) + f(b)] + h \sum_{i=1}^{r-1} f(t_i)$$

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Error of the Composite Trapezoidal Rule $\eta_i \in (t_{i-1}, t_i)$

Recall the result for one application of the trapezoidal rule:

$$\int_{t_{i-1}}^{t_i} f(x) dx - \underbrace{\frac{h}{2}[f(t_{i-1}) + f(t_i)]}_{I_{\text{trap}}} = \underbrace{-\frac{f''(\eta_i)}{12}h^3}_{\text{error}}$$

we have

$$\int_a^b f(x) dx = \underbrace{\sum_{i=1}^r \frac{h}{2}[f(t_{i-1}) + f(t_i)]}_{\text{composite trapezoid rule}} - \sum_{i=1}^r \frac{f''(\eta_i)}{12} h^3$$

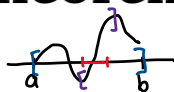
Assuming $f''(x)$ continuous on $[a, b]$, we have $\min_{x \in [a, b]} f''(x) \leq f''(\eta_i) \leq \max_{x \in [a, b]} f''(x)$

$$\min_{x \in [a, b]} f''(x) \leq \frac{1}{r} \sum_{i=1}^r f''(\eta_i) \leq \max_{x \in [a, b]} f''(x)$$

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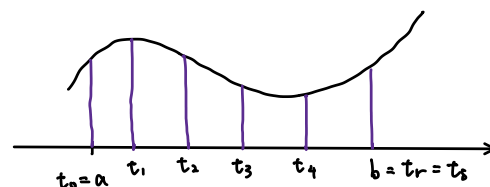
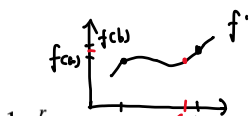
Intermediate Value Theorem

We will need to apply the Intermediate Value Theorem:



Let f be a continuous function on $[a, b]$. Then for each $c \in (f(a), f(b))$, there exists some $\xi \in (a, b)$ such that

$$f(\xi) = c$$



Let's apply this to f'' for the appropriately bounded value $\frac{1}{r} \sum_{i=1}^r f''(\eta_i)$:

$\exists \mu \in (a, b)$ s.t. $f''(\mu) = \frac{1}{r} \sum_{i=1}^r f''(\eta_i)$

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Error of the Composite Trapezoidal Rule Cont.

In conclusion, this gives us an error of

$$-\sum_{i=1}^r \frac{f''(\eta_i)}{12} h^3 = -\frac{1}{12} \left[\frac{1}{r} \sum_{i=1}^r f''(\eta_i) \right] r \cdot h \cdot h^2$$

$$= \frac{-f''(\mu)}{12} (b-a) h^2$$

where we used $b-a = h \cdot r$

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Composite Simpson and Midpoint Rules

A similar procedure for the other two quadrature rules we have studied gives us:

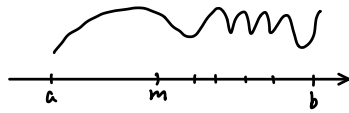
Midpoint: $\frac{f''(\mu)}{24} (b-a) h^2$

$\mu \in (a, b)$

Simpson's rule: $\frac{f''(\mu)}{180} (b-a) h^4$

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Adaptive Quadrature

Let's consider the following more general procedure:

Given a function $f(x)$ on $[a, b]$ and an absolute error tolerance ϵ_{tol} , find a quadrature (numerical integration) rule Q such that

$$|Q - I_f| \leq \epsilon_{tol}, \text{ where } I_f \triangleq \int_a^b f(x) dx$$

Adaptive Simpson's Rule

We are going to implement the search for this quadrature rule Q **recursively**. But first, let's introduce some notation:

Denote $h \triangleq b-a$ Simpson's rule error

$$\text{Next, } I_f = \int_a^b f(x) dx = S(a,b) + E(a,b)$$

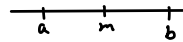
$$\text{where } S(a,b) = \frac{h}{6} [f(a) + 4f(\frac{a+b}{2}) + f(b)]$$

$$E(a,b) = -\frac{1}{90} (\frac{h}{2})^5 f^{(4)}(\xi), \quad \xi \in [a,b]$$

Further simplify: $S_1 \triangleq S(a,b)$, $E_1 \triangleq E(a,b)$

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Divide and Conquer



Let $m = (a+b)/2$ and apply Simpson's rule on $[a, m]$ and $[m, b]$:

$$I_f = \int_a^b f(x) dx = \underbrace{S(a,m) + S(m,b)}_{S_2} + \underbrace{E(a,m) + E(m,b)}_{E_2}$$

→ We can easily compute S_1 and S_2 (Simpson's rule)

→ How do we estimate errors E_1 and E_2 ? If $f^{(4)}$ does not vary "too much" on $[a, b]$, then: $E(a,m) = -\frac{1}{90} (\frac{h}{2})^5 f^{(4)}(\xi_1) = \frac{1}{32} [-\frac{1}{90} (\frac{h}{2})^5 f^{(4)}(\xi_1')]$,
 $\xi_1 \in [a, m]$
 $\approx \frac{1}{32} [-\frac{1}{90} (\frac{h}{2})^5 f^{(4)}(\xi)]$
 $= \frac{1}{32} E_1$

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Divide and Conquer

By a similar argument, $E(m,b) \approx \frac{1}{32} E_1$. Therefore,

$$E_2 = E(a,m) + E(m,b) \approx 2 \frac{1}{32} E_1 = \frac{1}{16} E_1$$

$$\text{From } I_f = S_1 + E_1 = S_2 + E_2$$

$$\begin{aligned} S_1 - S_2 &= E_2 - E_1 \approx E_2 - \frac{1}{16} E_2 \\ &= -\frac{15}{16} E_2 \end{aligned}$$

$$\Rightarrow E_2 \approx \frac{1}{15} (S_2 - S_1)$$

↳ we have an asymptotic expression for the error of our

→ we have an approximate expression for the error of the quadrature rule

$$\text{Finally, } I_1 = \int_a^b f(x) dx = S_2 + E_2 \approx S_2 + \frac{S_2 - S_1}{15}$$

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上一节课我们知道了相比于trapezoidal rule更好用的Simpson's rule，这节课我们讲一下 composition rule。如果说之前我们是用拟合多项式的阶数来优化积分，那么现在我们就是用更小的区间来做积分，减小误差。因为我们知道积分的误差和区间(b-a)^k成正比

我们推导了三种主要复合法则的**总误差公式**。关键结论是误差与步长 h 的几次方成正比。

- **复合梯形法则 (Composite Trapezoidal):**

- 推导逻辑：将所有小区间的误差 $-\frac{h^3}{12} f''(\eta_i)$ 加起来。
- 利用平均值定理，将求和项 $\sum f''$ 简化为 $f''(\mu)$ 。
- **最终误差**： $O(h^2)$ 。公式为 $-\frac{(b-a)}{12} h^2 f''(\mu)$ 。

- **复合中点法则 (Composite Midpoint):**

- **最终误差**： $O(h^2)$ 。公式为 $\frac{(b-a)}{24} h^2 f''(\mu)$ 。
- **对比**：精度也是二阶的，但系数是 $1/24$ ，比梯形法则 ($1/12$) 小一半，通常更精确。

- **复合辛普森法则 (Composite Simpson's):**

- **最终误差**： $O(h^4)$ 。公式为 $\frac{(b-a)}{180} h^4 f^{(4)}(\mu)$ 。
- **对比**：这是一个巨大的飞跃。步长减半，误差减小 16 倍，是高精度计算的首选。

关于你问的 μ (mu) 到底是什么，这是一个非常好的数学概念问题。

简单来说： μ 是在积分区间 (a, b) 内的某一个具体的（但通常未知的）点。

这个点之所以存在，是因为数学上的**中值定理 (Intermediate Value Theorem)**。下面我带你一步步拆解这个过程，你就明白了：

1. 为什么会有 μ ？（从“很多个点”变成“一个点”）

当我们做复合积分时，我们其实是把一大堆小区间（一共 N 个）的误差加起来：

$$\text{总误差} = \sum_{i=1}^N (\text{第 } i \text{ 个小梯形的误差})$$

对于每一个小区间，误差都取决于那个特定小区间里的某一点 ξ_i 的二阶导数 $f''(\xi_i)$ 。所以总误差写出来其实是这样的一大串求和：

$$\text{总误差} = -\frac{h^3}{12} [f''(\xi_1) + f''(\xi_2) + \cdots + f''(\xi_N)]$$

这就很麻烦，因为这里有 N 个不同的二阶导数值。我们想把公式简化。

2. 引入“平均值”概念

利用数学变换（因为 $h = \frac{b-a}{N}$ ），我们可以把上面的式子改写成：

$$\text{总误差} = -\frac{h^2(b-a)}{12} \times \underbrace{\left[\frac{f''(\xi_1) + f''(\xi_2) + \cdots + f''(\xi_N)}{N} \right]}_{\text{二阶导数的平均值}}$$

看方括号里的部分，这实际上就是所有这些小区间里二阶导数的**算术平均值**。

- μ 是什么？它是区间 (a, b) 里的某一个点。
- 它代表什么？ $f''(\mu)$ 代表了函数 $f(x)$ 在整个积分区间上二阶导数的平均表现 (Average behavior)。
- 注意：
 - 梯形法则里的 μ 和中点法则里的 μ 不是同一个点（就像你去程和回程达到平均速度的地点可能不同），但它们都位于 (a, b) 之间。
 - 对于辛普森法则， μ 对应的则是四阶导数 $f^{(4)}$ 的平均值点。

Adaptive Quadrature 是一个不需要手动设置步长 h ，而是通过设置 tolerance error ϵ_{tol} 的方式让算法自己去寻找满足 $|Q - I_f| < \epsilon_{\text{tol}}$ 的方法。

1. 先在整个区间上用一次 Simpson rule $[a, b]$ 得到 S_1
2. 用中点 m 把 $[a, b]$ 区分为两个区间，在这两个区间上分别使用 Simpson rule

在不知道真实值 I_f 的情况下，怎么估计 error?

我们最终推导出了公式， $E_2 = \frac{1}{15}(S_2 - S_1)$